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Maximizing Drilling Efficiency: Leveraging AI Technology to Surpass Drilling Technical Limit in Brownfield-Fast-Drilling Environment

Bastian Andoni, Andre Wiharja, Bobby Wijaya Hendarno, Disha Fitri Azalia, Muhammad Ihsanul Amal, and Crisa Agriawan, Pertamina Hulu Mahakam; Mabel Lim, Febrian Arie Wibowo, Ni Made Wilasari, and Gerardo Ivan Hernandez, SLB

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Abstract

The objective of the operator is to enhance drilling performance and efficiency by implementing a new technology to achieve consistent Rate of Penetration (ROP). The goal is to consistently drill one mile within the first 24 hours of drilling. Multiple approaches have been tried in the past but achieving consistency was a challenging task, in this case the operator embedding new approaches by utilizing Artificial Intelligence (AI) to achieve consistency in drilling operation.

This new approach brings the AI to the rig-site and is tasked to advise driller achieving consistent ROP within safe operation window. This technology incorporated Edge computing technology and human interaction through Human Machine Interface (HMI) to enable driller-AI interaction. Respective formation, rig and tools limitations for drilling operational safety, parameter constraints are set prior drilling each well. Real-time drilling data is aggregated and ingested by the AI at the Edge considering all the prior information and constraints to recommended drilling parameters while honoring operational safety. The continuous feedback loop audits the optimization, providing consistent output for optimized drilling parameters.

Deploying AI Edge advisory is expected to significantly enhance drilling efficiency and safety. The system's AI-driven insights optimize the rate of penetration (ROP) and mitigate risks such as Shock & Vibration (S&V), Stick slip (S&S), and dysfunctional mitigation, leading to smoother operations and reduced downtime. System deployment will improve and accelerate decision-making due to high-quality visualizations and real-time data analysis, which help drillers respond promptly to changing conditions. Integration of system with existing rig hardware ensures minimal disruption to existing rig set-up, establishing smooth transition of working environment for the rig crew. Additionally, training and workshops help teams overcome apprehensions about digital enhancement, fostering a positive attitude and smoother adoption of drilling practices for the future.

In conclusion, the deployment of AI at the Edge, deployed at rig-site, results in more efficient, safer, and cost-effective drilling operations. The system's adaptability to real-time conditions and its ability to provide actionable insights ensure consistent performance improvements. This leads to a more competent and confident drilling team, ultimately driving better operational outcomes and higher productivity

This technology uses proprietary AI with the Detourney model, deployed and performed at the Edge for seamless adoption. It dynamically reacts to real-time drilling conditions, optimizing processes based on downhole observations. The smart AI Planner autonomously navigates workflows via a real-time feed. It embeds hazard identification such as high torque, stick slip, shock vibration and also provides advisory to mitigate risks when detected, then continues normal drilling once risks are mitigated.

Introduction

Delta Mahakam block located in East Kalimantan is one of the oldest oil and gas fields in Indonesia. The first production was 50 years ago in 1970 from one of the fields in Delta Mahakam block, Bekapai field. Bekapai field has since been heavily developed. When current operator took over in 2018, the field has already been considered as a brownfield facing massive decline, with low drilling activities prior to the block handover. Fig. 1 shows the continuous decline trend from 2010 until 2018, with a drastic decline towards 2018.

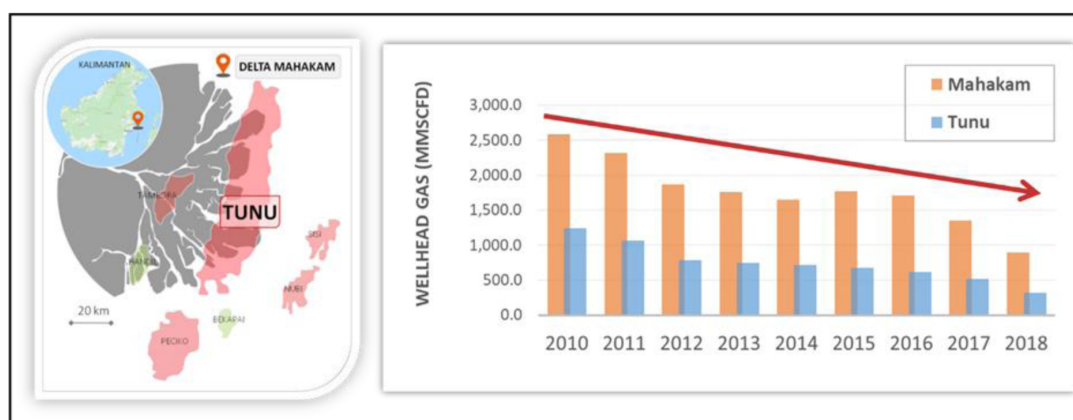


Figure 1—Delta Mahakam block production decline trend (Sayogyo et al., 2022)

In addition to the block massive decline rate, the geological setting consisting of reservoirs pocket (sand lenses), are getting more challenging to reach. In order to meet production targets, the operator strategized to increase the number of wells, focusing on targeting the reservoir pocket. In parallel, heavy exercises on well costs spendings was also executed. Fig. 2 shows the profile of average well budgets against the actual well cost over the years.

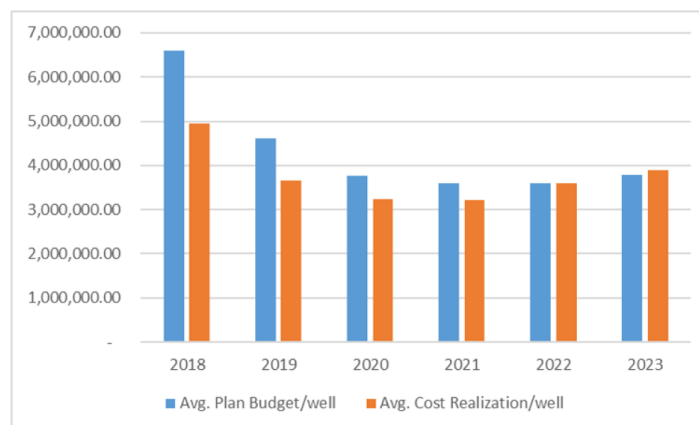


Figure 2—Average well cost (budget vs actual) in Delta Mahakam block

Multiple approaches have been deployed to reduce the drilling cost to support the economic development of the field. Drilling operational efficiencies practices were strongly enforced by the Mahakam block

operators, with continuous improvement methods. Hadistira et al. (2021) described some of the initiatives that have been done in the fields, including co-activity operation (batch drilling, rigless CP driving, and CWH installation, CP survey while CP clean out, no WOC on surface casing cementing, etc.), Mastering drilling practice (shallow gas best practice, enhanced drilling parameter, bit selection strategy, and fast string connection). All the optimization has made the drilling environment become factory-drilling with a fast operation and is considered to have reached the operational technical limit. However marginally, the Mahakam Block owners continue to explore technologies for pushing the technical limits. In this paper, artificial intelligence Edge technology was explored.

Challenges

Optimizing drilling operations performance and pushing the consistent technical limits in Delta Mahakam fields is a challenge. The current development wells have reached maturity, where finding new and economic targets are with high technical challenge such as low or marginal reserve (< 1 bcf). In addition, due to the downturn of oil and gas in 2014, completing these marginal wells with prior approaches makes the well uneconomical. Fig. 3 shows the historical number of Field T Main Zone (TMZ) and Field T Shallow Zone (TSZ) number of drilled wells and well duration from the early 1980s until 2019. The well duration decreases significantly as a result of operation optimization that has been continuously happening over the years of operation. The TSZ just started in 2015, which indicated that the shallow gas reservoir was not becoming the production interest previously, and as the current reservoir becomes depleted and marginal, a way to produce hydrocarbon from the shallow zone must be discovered.

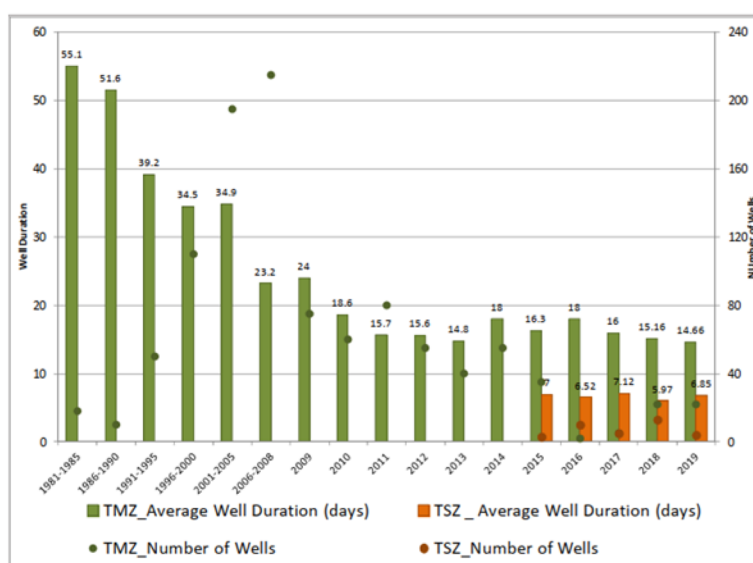


Figure 3—Average well duration and numbers of wells evolution

As the reservoir in Delta Mahakam keeps being produced, the available reservoir targets become fewer. Additionally, the size of the reservoir continues to decline. Building a new platform becomes the last option considering the economic benefit of it. As more wells are being developed with existing slots and surface locations, added difficulties and complexities with trajectory designs (complex profile, high inclination, long departure). This inevitably causes another added challenge on hole cleaning effectiveness during drilling and tripping (Fig.4).

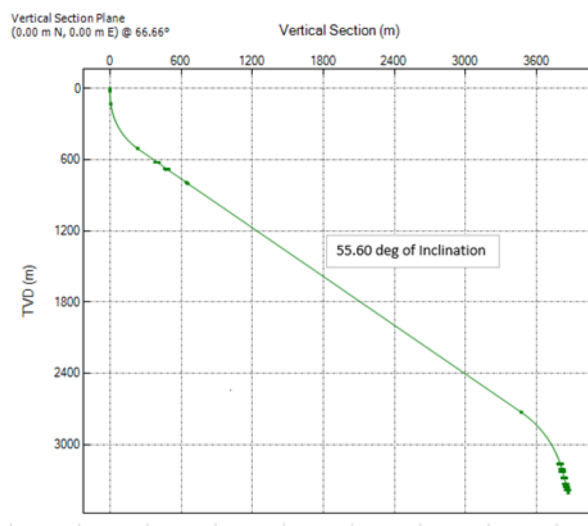


Figure 4—Typical well with high inclination and long departure, required higher flow rates for hole cleaning

There are several challenges on directional drilling activities. The first challenge is dogleg severity (DLS) capability to achieve the trajectory target. It was recognized that steering difficulties were seen in the T field using the elastomer-based push-the-bit Rotary Steerable System (RSS) BHA, where a build interval in the 8.5in. hole section is present in the main zone wells (deeper than 1900 m TVD). One of the solutions was to control the parameters (lower RPM, reduce ROP) to increase the effectiveness of the RSS pads. However, it impacted the drilling duration negatively (higher rig time). A technological breakthrough needed to be implemented to solve this issue challenge (Fig.5).

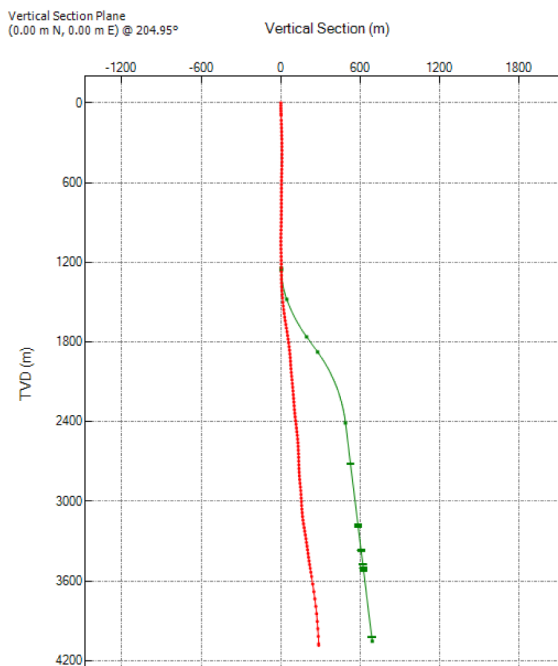
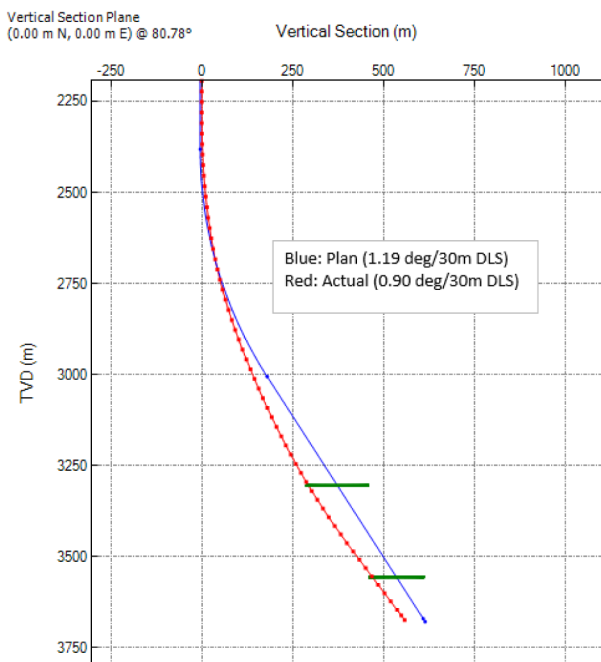


Figure 5—Left: steering issue encountered in the T field using elastomer-based RSS technology; Right: Typical well in TM field with S-Type profile

Another challenge that is faced in Delta Mahakam development is the sharp pressure ramp-up. This pressure ramp up begins at approximately 3,400 mTVDSS and deeper. In some wells, the reservoir to be targeted is located very close to this ramp-up zone. This causes difficulties and complexities in determining

the final well depth accurately. Drilling through this zone can cause a serious well-controlled situation which must be avoided. Drill pipe torque limitations are expected, with need of having a closer-to-the-bit and more accurate formation evaluation. The Mahakam Operators continues to explore ways of technological breakthrough to safely drill this section.

The optimization in the brownfield fast drilling environment is not only on the productive drilling operation but also on the invisible lost time (ILT). ILT is an unseen non-productive time that is conventionally considered part of productive activity. Although optimizing (or even eliminating) ILT is often considered minor with a very small potential of rig time saving, the consistent implementation might lead to a significant cumulative gain. Some potential ILT reductions were identified, mainly covering BHA surface handling (make-up and lay down) time, pipe connection time, and drilling performance inconsistency (depending on driller's level of experience and expertise). All these were sought as potential marginal rig time-saving gains that were worth pursuing.

With an increasing level of drilling difficulty, the expertise and experience of a driller play a crucial role in the performance of a drilling operation. Reliance on the driller's capabilities causes drilling performance in Mahakam to be inconsistent. There is an indication that drilling performance in Mahakam delta area is quite influenced by the driller operating the brake at that time.

As can be seen in Fig. 6 and Fig. 7 which compare the drilling performance in field T using two different rigs. Fig. 6 shows average ROP performance in field T with rig R, while Fig. 7 shows average ROP performance in field T with rig Y. It can be seen that the average performance of rig Y is better than the performance of rig R. Even though there are many factors affecting the drilling performance such as well trajectory, formation drilled, downhole hazard (loss prone interval, fault, high pressure formation, etc), human factor also plays a quite significant contribution in terms of achieving a constant maximum drilling performance.

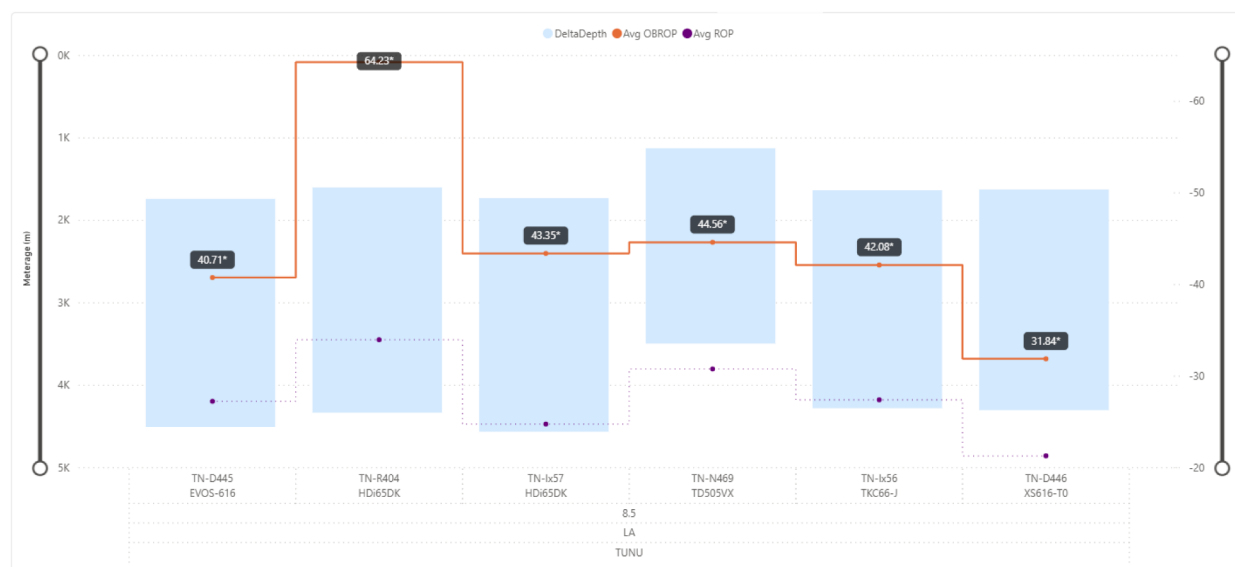


Figure 6—Average On Bottom ROP in field T with rig R

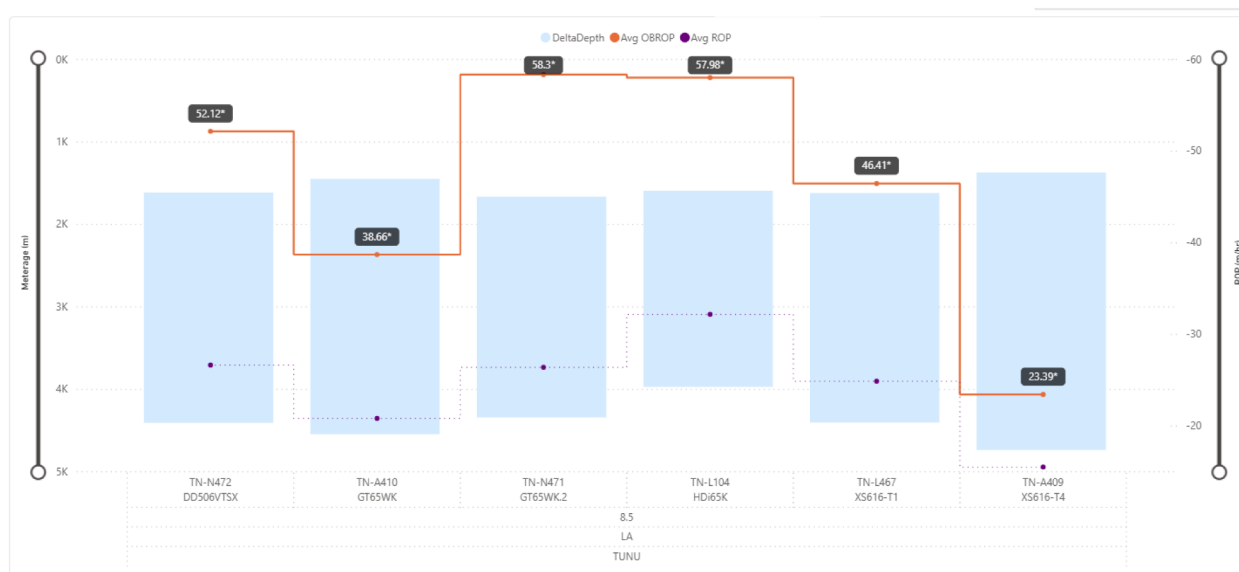


Figure 7—Average On Bottom ROP in field T with rig Y

Utilization of a state-of-the-art artificial intelligence technology which can advise most optimum parameters during drilling, especially in a deeper hole interval, can seem to be one of the feasible solutions to standardize the drilling performance for all drilling rigs which further can eliminate the inconsistency in drilling performance.

Methodology

Technology that can help boost efficiency is crucial to solving all the problems. The management and optimization of drilling operations in the oil and gas sector require a thorough strategy. It entails combining several technologies and procedures to enhance the efficacy, safety, and efficiency of drilling operations. This concept is referred to as Digital Drilling, and it encompasses the following key components:

1. **Integration and Analysis of Data:** Digital drilling systems gather and combine data from a number of sources, such as sensors, drilling parameters, downhole tool's readings, and geological information. Advanced data analysis and algorithms are then applied to provide actionable insights, such as parameter recommendations to optimize the rate of penetration.
2. **Optimization Algorithms:** Optimization algorithms are used to enhance the efficiency of drilling operations such as ROP optimization and drill string integrity monitoring. These algorithms act as intelligent tools that assist drillers in evaluating and determining the most suitable parameters for various drilling conditions. By providing data-driven recommendations, they help standardize the decision-making process, reducing reliance on individual experience levels across different crews. This not only bridges the gap between varying levels of expertise but also ensures greater consistency in applying optimal parameters throughout the well construction process, ultimately improving performance and reducing invisible lost time.
3. **Safety and Compliance:** While efficiency improvement is the goal, technology must adhere to safety factors. The digital drilling technologies guarantee that drilling activities adhere to safety considerations and requirements.
4. **Collaboration and Communication:** During operational execution, the system is additionally outfitted with the ability to integrate and collaborate across various teams involved in the drilling process.

System Deployment

The edge box device and the human machine interface (HMI) are two essential elements that must be implemented in the system deployment. Hardware elements that process data at a network's edge include sensors, gateways, routers, and Internet of Things devices. In edge computing, where data is processed closer to its source to minimize latency, enhance real-time responsiveness, and maximize bandwidth utilization, these devices are essential. HMI, on the other hand, is a technology that enables people to communicate with machines and systems, usually in an operational or industrial setting. HMIs are the interface that allows users to track procedures and do real-time data analysis. It includes an easy-to-use interface that displays data from the system.

The Human-Machine Interface (HMI) is typically deployed within the driller's cabin to provide real-time visualization and control capabilities. In contrast, the edge computing device is strategically located within a designated safe zone, usually in an area compliant with hazardous area classification standards to ensure compliance with safety regulations and to facilitate local data processing. For the execution of the AI-driven analytics workflow, the system requires continuous acquisition of surface drilling parameters. These parameters, such as Weight on Bit (WOB), rotary speed (RPM), torque, standpipe pressure (SPP), and rate of penetration (ROP) are transmitted via a WITS (Wellsite Information Transfer Specification) interface from the Measurement Logging Unit (MLU), which acts as a centralized data aggregation point (Fig.8).



Figure 8—System Deployment in the rig site

Simultaneously, downhole telemetry data acquired from Measurement While Drilling (MWD) and Directional Drilling (DD) tools are critical for maintaining drill string integrity and operational safety. This includes but is not limited to directional survey data (inclination and azimuth), shock and vibration measurements, and stick-slip indicators. These downhole parameters enable advanced condition monitoring and predictive diagnostics to mitigate the risk of tool failure or wellbore instability during drilling operations.

System Workflow

ROP Optimization. In the past, maximizing the Rate of Penetration (ROP) primarily involved pushing drilling parameters such as Weight on Bit (WOB), Revolutions Per Minute (RPM), and torque to their limits, within the constraints of formation strength and downhole tool capabilities. This process depended heavily on the skill and experience of the driller and directional driller.

As described by Barajas et al. (2023) in their study, driller's experience significantly affects the performance and stability of autodriller systems. Inconsistent set point management, lack of awareness of control loop behavior, and delayed response to surface data can all lead to increased dysfunction, including stick-slip, motor stalls, high torque variance, and ROP fluctuations.

Experienced drillers who understand which loop is in control and recognize signs of dysfunction such as overshoot patterns or repeated oscillations are better equipped to mitigate it by adjusting setpoints or switching control modes. With the advent of drilling automation systems, this approach has evolved. Automation now allows for real-time adjustment of drilling parameters to determine the most effective combination that delivers maximum ROP while staying within defined operational limits.

A key component of this system is the AutoROP engine, which uses both surface and downhole data to make intelligent, real-time recommendations. By continuously analyzing real-time drilling conditions, the system builds a dynamic model of the formation, estimating formation strength and predicting the optimal combination of drilling parameters—such as RPM, WOB, ROP, differential pressure, and torque—for each interval. It also calculates the theoretical ROP at the bit, accounting for the dynamic response delay between surface and downhole parameters. This enables the driller to apply precise setpoints that maximize performance within operational constraints while minimizing the risk of drilling dysfunctions.

The ROP optimization engine operates as illustrated in Figure 9. Initially, it receives operating limits specified by the user. In Step 1, the system ingests real-time drilling sensor data. Step 2 employs a drill string compression model to convert surface measurements into downhole parameters, calculating downhole weight on bit (WOB), downhole torque, and rate of penetration (ROP) at the bit. Utilizing these downhole parameters, Step 3 constructs a dynamic formation model that outputs optimized drilling parameters, including recommended WOB, rotary speed (RPM), and theoretical ROP. Step 4 incorporates a proprietary model validation and update mechanism, which continuously assesses the accuracy of the formation model fit against actual downhole conditions; upon detecting model discrepancies, it recalibrates and generates updated formation models to ensure optimal performance under evolving drilling conditions.

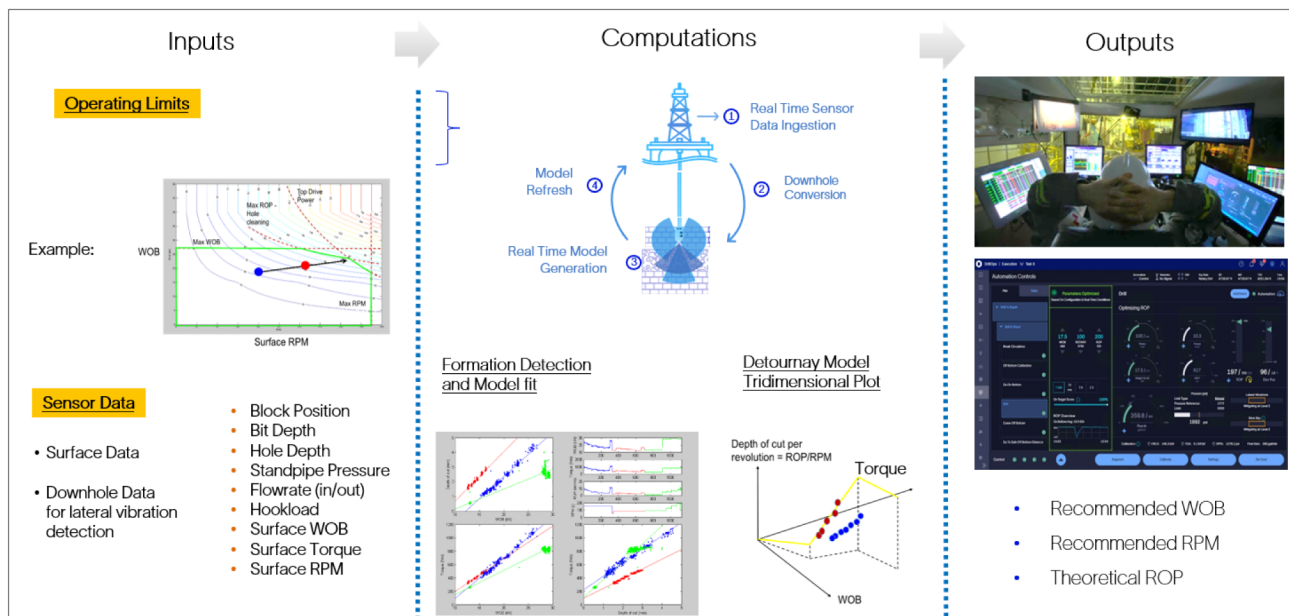


Figure 9—ROP Optimization Workflow

Drilling Dysfunction Mitigation. While the ROP optimization engine minimizes the risk of drilling dysfunctions caused by improper surface parameter management, downhole tools may still experience dysfunctions such as stick-slip and vibrations due to formation variability and mechanical interactions.

These dysfunctions can cause serious issues during drilling, including premature tool wear, failures, and increased non-productive time. Traditionally, identifying and mitigating these problems was the responsibility of drillers, directional drillers, and MWD (Measurement While Drilling) engineers. However, delays in detection or inconsistent responses often led to less-than-optimal results.

Modern drilling automation systems address these challenges by incorporating built-in logic to detect dysfunctions in real time and trigger appropriate mitigation measures. These mitigation protocols are preconfigured based on the operator's standard operating procedures and industry best practices, ensuring a consistent and reliable response.

The system automatically adjusts key parameters, mainly WOB and RPM, to counter the dysfunctions effectively. Once the issue is resolved, the system quickly transitions back to optimizing ROP, maintaining drilling efficiency with minimal disruption. This automated, closed-loop approach ensures a balance between mitigating dysfunctions and maximizing overall drilling performance as illustrated in Fig.10.

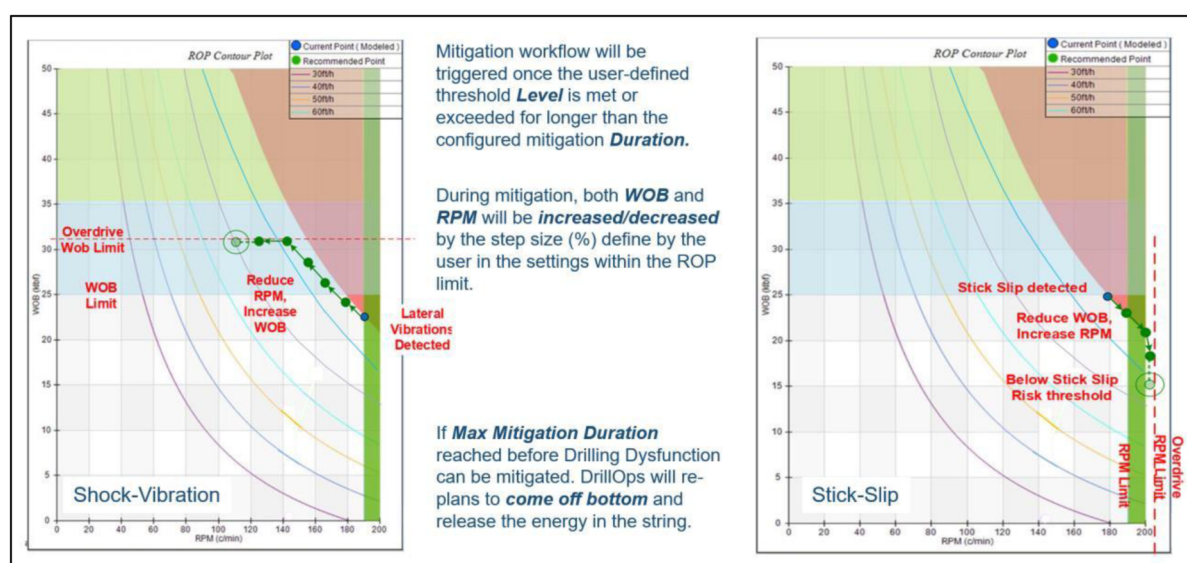


Figure 10—Dysfunction Mitigation Workflow

Drilling conditions can vary significantly between stands, necessitating an intelligent orchestrator capable of dynamically prioritizing between optimizing rate of penetration (ROP) and initiating mitigation procedures for drilling dysfunctions. This orchestration leverages artificial intelligence through the use of Planning Definition Domain Language (PDDL). PDDL is a formal language designed to represent planning problems, enabling an automated planner to generate a sequence of actions to achieve specific objectives. It encodes the domain rules, problem constraints, and available actions, allowing the planner to derive valid and optimized operational plans. A practical analogy for PDDL application is autonomous vehicle navigation, where the system must plan a route from Point A to Point B while considering dynamic constraints such as road closures or traffic restrictions.

In the drilling context, the AI planner utilizes PDDL to enable reasoning over complex, hierarchical tasks. The planner receives input from various domain-specific engines, alongside its knowledge of the rig's current state and the user-defined objective, typically drilling to the target depth. It then computes the most optimal sequence of operations to achieve this goal. For example, when the AutoIntegrity domain engine (as shown in Fig. 11) detects a risk of vibrations, it triggers an alert to the AI planner. The planner subsequently schedules and orchestrates the necessary mitigation actions. Once the vibration issue is resolved, control is returned to the ROP optimization process until the drilling stand is completed, and the target depth is reached.

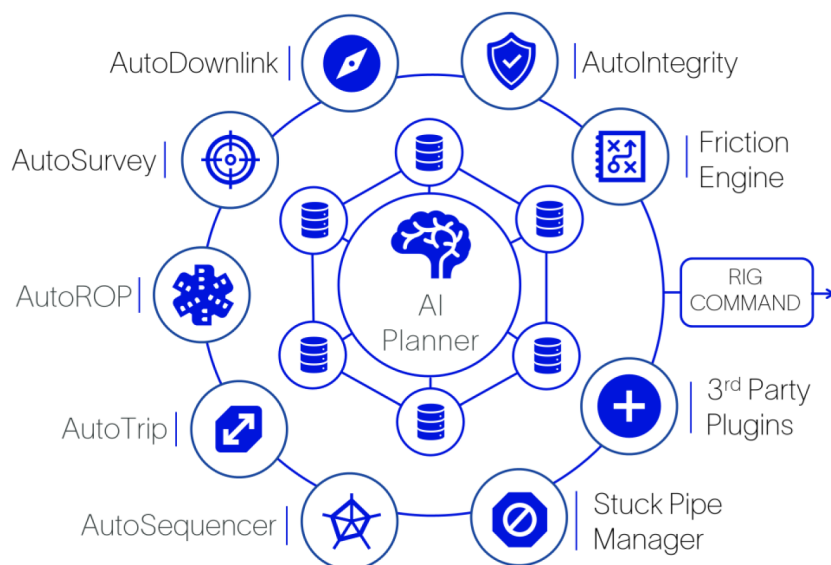


Figure 11—AI Planner Design Methodology

Case Study from Delta Mahakam

A number of wells and well sections on Rig R have incorporated AI-based drilling optimization solutions. As part of this initiative, multiple wells across multiple fields were drilled utilizing AI-driven workflows aimed at enhancing Rate of Penetration (ROP) performance. This report focuses specifically on the implementation and performance outcomes of the AI solution in the TM field.

Within the TM field, two development wells, designated TM-1A and TM-5A, were drilled utilizing the AI-assisted approach. Both wells are sidetracked with light well architecture classification. The planned trajectory for each well conforms to a 3D S-curve profile, characterized by a maximum inclination of approximately 27 degrees. The anticipated dogleg severity (DLS) for both builds and turn sections ranges from 2.5°/30m to 4.0°/30m (Fig.12).

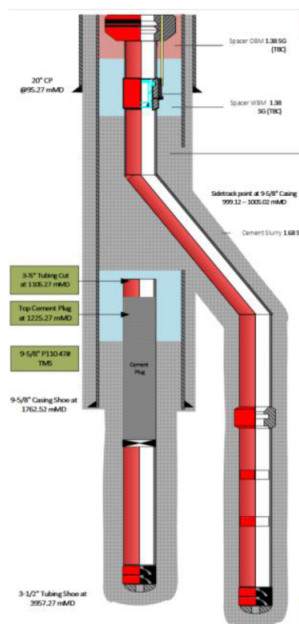


Figure 12—Typical Well Diagram in TM Field

Each well is designed to be drilled through a single section: an 8.5-inch hole interval, which will be completed using a tubingless completion strategy. This configuration is consistent with field development objectives focused on cost-efficiency and simplified well architecture, while still enabling real-time drilling optimization through AI integration.

Field Deployment

Field deployment and pre-implementation of the system involve several critical steps to ensure successful integration and operation as seen on [Fig.13](#).

1. Planning and Preparation

Before deploying the system in the field, thorough planning and preparation are essential. This includes assessing the specific needs and requirements, such as rig survey, measuring the needs of miscellaneous equipment and accessories to install the hardware (Edge Box), and developing a detailed implementation plan. Coordination and discussion with all stakeholders, including drilling engineers and rig personnel to ensure alignment and readiness.

2. Equipment Installation and Testing

The next step after planning and preparation is to install and test the hardware in the field. This includes setting up the equipment, connecting the edge box to the HMI, establishing a connection to the company man viewer, configuring WITS, setting up the data transmission system, and establishing the communication network. The system hardware must be carefully tested to ensure accurate data collection and reliable operation.

3. Data collection, QAQC, and System Tuning

Gathering information from various sources, including surface sensors, downhole tools, and all other contextual information. It also needs data validation through QAQC process, checking the data for accuracy and consistency. It also includes verifying that data collected from surface sensors and downhole tool measurements are within the expected ranges, and any discrepancies or errors are identified to ensure the integrity of the data.

4. Hypercare (Training and Support)

Training is a critical component of field deployment. Operators, drillers, and other personnel must be trained in how to use the system effectively. This includes understanding the system's capabilities, interpreting data and recommendations, and responding to the advisory. Ongoing support and troubleshooting are also necessary to address any issues that may arise during operation.

5. Continuous Monitoring and Optimization

After deployment, continuous monitoring and optimization are essential to ensure the system operates at peak performance. This involves regularly reviewing data and implementing improvements based on feedback and performance metrics. The goal is to achieve consistent and efficient drilling operations by optimizing ROP while minimizing downtime and maximizing safety.

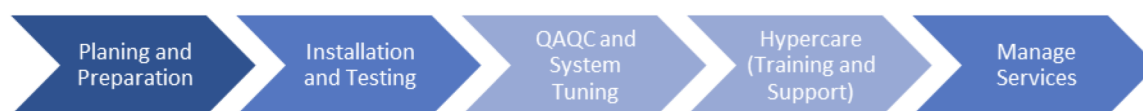


Figure 13—Field Deployment Process and Workflow

Performance Benchmarking

Performance benchmarking is performed as a systematic process of comparing the average On Bottom ROP on group of wells with the same rig, same field, the same BHA configuration, and well profile. The benchmarking process was performed to understand the performance on each well against predefined metrics and historical data. It establishes a baseline for comparing performance across well, field, and rigs.

The ROP benchmarking in TM field was completed by classifying the BHA configurations that are frequently utilized in this field which is BHA RSS and BHA Motorized RSS. The well TM-1A is drilled by using RSS BHA, meanwhile well TM-5A is drilled by using motorized RSS BHA.

For BHA Motorized RSS, the baseline ROP benchmark was approximately around 65 m/hr, meanwhile for BHA RSS the ROP benchmark is around 69 m/hr.

BHA Motorized RSS. Based on the ROP performance from TM wells which has been clustered into the same BHA configuration, BHA Motorized RSS, it is shown that the On Bottom ROP average is 65 m/hr, based on the performance analysis from offset wells, TM-A, TM-B, TM-C, TM-D, and TM-E. The ROP performance is good in the shallow zone, the average On Bottom ROP from interval 1000-3000m TVD can vary between 80-120 m/hr, and it will be significantly dropped after 3000m TVD to be around 30-80 m/hr, the drop even prominently after 3500m TVD to be below 30 m/hr in average. It indicates that the compressive strength and lithology become harder as we get deeper, with no improvement in ROP performance in the deeper section of the formation (Fig.14).

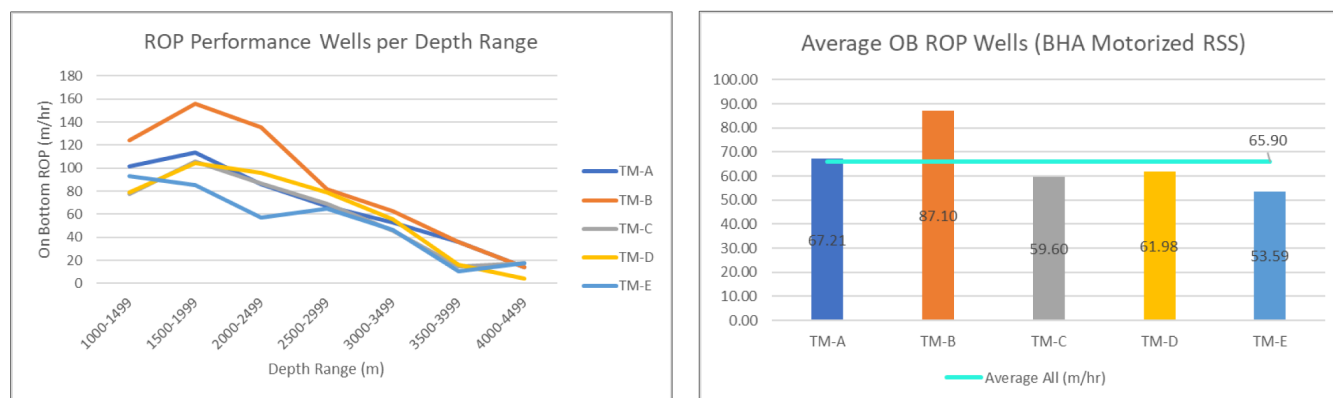


Figure 14—On Bottom ROP Wells Performance and Benchmark with BHA Motorized RSS

BHA RSS. Meanwhile, for BHA RSS based on the offset well comparison in TM-F, TM-G, TM-H, TM-I, TM-J the average on bottom ROP is around 69 m/hr. Almost the same with Motorized RSS, the trend of on bottom ROP is declining as soon as the well getting deeper. From 1000-3000m TVD average on bottom ROP is vary between 60-120 m/hr and it will significantly drop after 3500m TVD to 30-50 m/hr (Fig.15).

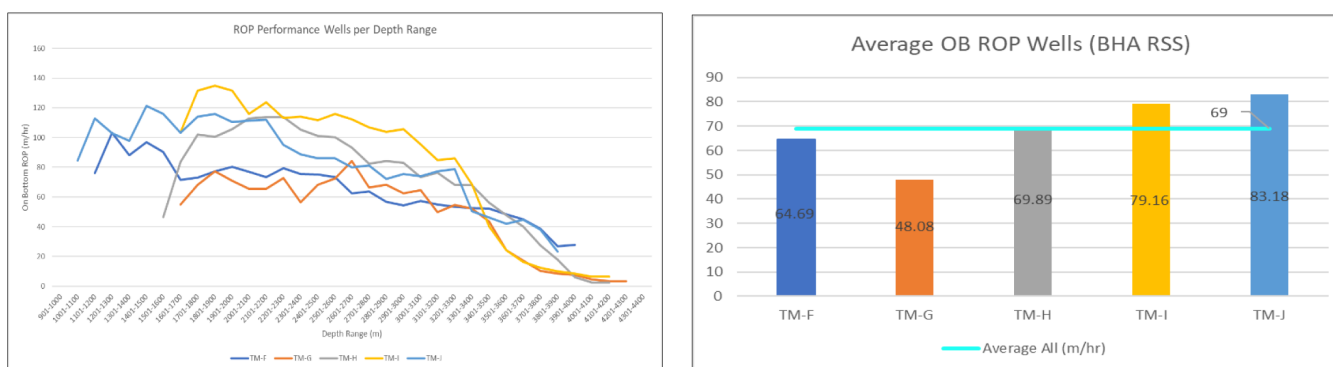


Figure 15—On Bottom ROP Wells Performance and Benchmark with BHA RSS

Technical Limit Assessment

A technical limit assessment is essential for improving well and project performance because it helps identify the maximum safe and efficient operational boundaries. By understanding these limits, teams can optimize drilling parameters, prevent equipment failures, and reduce non-productive time. It also ensures

safety standards are maintained while pushing operational capabilities, ultimately leading to enhanced efficiency, cost savings, and successful project execution.

As an input for the AI System, some of the parameter limits need to be defined, such as RPM, WOB, ROP, Flow Rate, Torque, and Differential Pressure. Below are a few key points to consider regarding the technical limit numbers (Fig.16).

Parameters	Limiting Factor
Weight on Bit	Torque and Drag Buckling
	Jar Placement
	Maximum allowable WOB for Bit
Surface RPM	Max allowable RPM of Top Drive
	Max allowable RPM for Motor/RSS
	Max allowable RPM for other downhole tools (reamer, well commander, PBL)
Flowrate	Rig Pump Pop-off pressure set
	Min and Max flowrate for DD/MLWD tools
	Min and Max flowrate for other downhole tools
ROP	RT/RM LWD data update rate
	Steering purpose
Surface Torque	Max allowable STOR for Top Drive
	Max and recommended make up TQ
Differential Pressure (if Motor BHA)	As per tool spec sheet if Motor is used

Figure 16—Technical Limiting Factor

A technical limit assessment has been conducted by integrating information from various sources, including rig capabilities, directional drilling considerations, tool specifications, and data acquisition requirements. For example, in the case of Weight on Bit (WOB), multiple limiting factors were considered, such as the bit's maximum WOB rating, jar placement, and torque and drag-induced buckling risks. These limiting factors must be thoroughly reviewed and agreed upon prior to the commencement of drilling operations. It is important to note that technical limits may vary across different sections and depth intervals, depending on specific downhole conditions and operational constraints.

Technical Limit Well TM-1A. TM-1A is using BHA RSS, hence the differential pressure technical limit is not applicable.

Table 1—Technical Limit for TM-1A

Parameter	Value	Unit
<i>Weight on Bit</i>	16	t
<i>Surface RPM</i>	180	rpm
<i>Flow Rate</i>	2350	lpm
<i>Surface Torque</i>	43	klbs.ft
<i>Rate of Penetration</i>	150	m/hr
<i>Differential Pressure (if Motor BHA)</i>	n/a	-

Technical Limit Well TM-5A. Well TM-5A is using BHA Motorized RSS, the technical limitation as follows.

Table 2—Technical Limit for TM-5A

Parameter	Value	Unit
<i>Weight on Bit</i>	16	t
<i>Surface RPM</i>	100 (curves section) 120 (tangent section)	rpm
<i>Flow Rate</i>	2250	lpm
<i>Surface Torque</i>	43	klbs.ft
<i>Rate of Penetration</i>	150	m/hr
<i>Differential Pressure (if Motor BHA)</i>	n/a	600 psi

Execution and Result

The 8.5-inch hole section of Well TM-5A was successfully drilled utilizing the AI-assisted drilling optimization system. Implementation of the AI workflow resulted in a measurable improvement in drilling efficiency, with the Rate of Penetration (ROP) increasing by approximately 6%, from a baseline average of 65 m/hr to 69 m/hr (Fig.17).

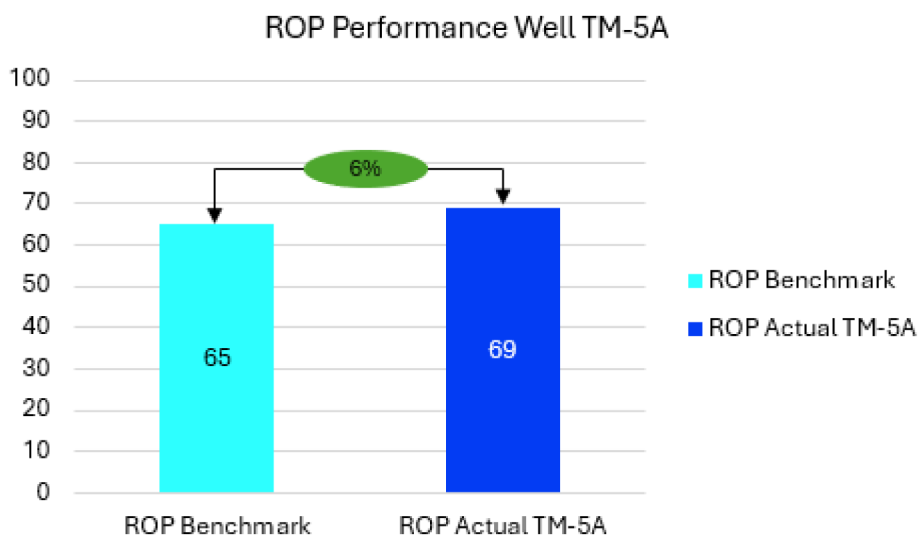


Figure 17—ROP Performance in TM-5A

Despite the performance gains, several operational constraints were encountered during this section. The primary limiting factors were related to Equivalent Circulating Density (ECD) management and the maximum allowable differential pressure. These parameters imposed restrictions that prevented the application of the originally planned upper operational thresholds across specific depth intervals.

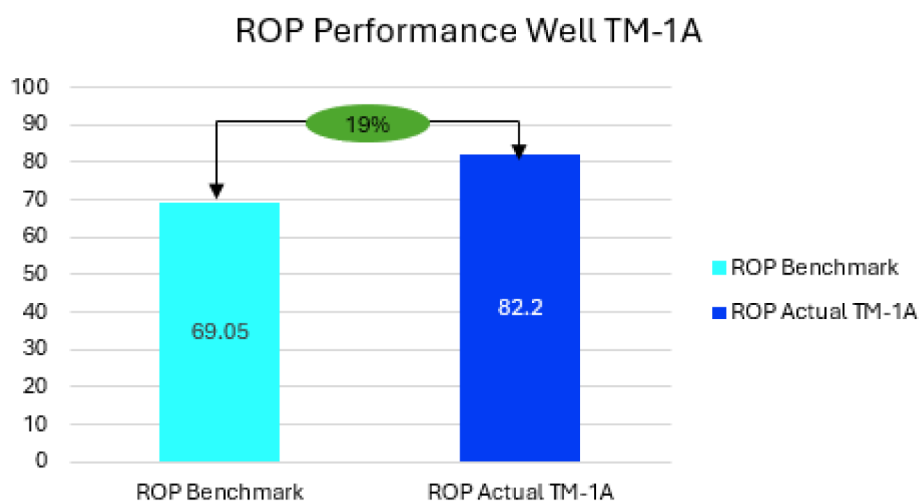
Under these conditions, the Rate of Penetration (ROP) was intentionally controlled to ensure safe operations. The operational focus shifted toward effective hole cleaning and extended circulation to mitigate ECD related risks. Once the ECD issue was resolved, ROP optimization resumed, and drilling continued with the objective of maximizing penetration rates.

To maintain wellbore integrity and ensure safe drilling practices, any adjustments to the predefined technical operating limits, whether driven by real-time conditions or optimization recommendation, were

subject to prior review and approval by the Drilling Supervisor. This collaborative decision-making process was essential for balancing performance objectives with operational risk mitigation.

For dysfunction mitigation, the trigger threshold was set at Level 3 severity. However, throughout the entire run, no high-severity events such as stick-slip, shock and vibration, or stringer encounters exceeding Level 3 were observed. This indicates that the drilling performance remained within manageable operational envelopes, and the system effectively maintained stability without triggering major dysfunction responses.

Meanwhile, in the second well, TM-1A, the Rate of Penetration (ROP) showed a substantial improvement of 19%, increasing from a benchmark of 69 m/hr to 82 m/hr (Fig.18). This mirrors the performance trend observed in the earlier of TM-5A well. Several limitations were encountered during drilling operations. These included high Equivalent Circulating Density (ECD), challenges in trajectory control, and erratic torque behavior. Under these conditions, drilling parameters were carefully managed by adjusting the Rate of Penetration (ROP) to optimize hole cleaning and controlling the Rotation Per Minute (RPM) to maintain trajectory accuracy and ensure drill string integrity. During these conditions, the ROP optimization mode was temporarily disabled to ensure that appropriate mitigation measures were in place, with a focus on effective hole cleaning and maintaining proper trajectory control.



In particular, one interval exhibited a tendency for inclination to drop, requiring ROP to be deliberately reduced and controlled within the range of 30–50 m/hr to maintain the planned well trajectory. Additionally, approaching Total Depth (TD), the well experienced pronounced erratic torque events. As a mitigation measure, both Weight on Bit (WOB) and ROP were proactively adjusted to minimize dysfunction and ensure operational stability.

During the drilling of this section, two stick-slip events were detected and effectively mitigated. The timely intervention not only enhanced drilling system stability but also contributed to preserving tool integrity and extending the service life of downhole equipment. By proactively addressing these dysfunctions, the operation avoided potential drill string damage and other unwanted events related to mechanical integrity.

In both wells where the AI solution was implemented, a clear improvement in drilling performance was observed. In the first well, a moderate ROP increase of 6% was achieved using a motorized RSS BHA. In contrast, the second well demonstrated a significantly higher ROP improvement of 19% with a fully rotary steerable system (RSS) BHA. This positive trend highlights the AI system's capability to support real-time decision-making by providing parameter optimization recommendations to the driller. As a result, the system played a key role in driving performance gains and helping achieve the overall objective of ROP enhancement. This uplift demonstrates not only the system's capability to optimize on-bottom drilling

parameters in real time but also its direct contribution to improving drilling efficiency, reducing operational time, and lowering cost per foot.

Conclusion

The implementation of the AI solution for drilling optimization system has demonstrated measurable performance benefits across both wells. While the first well achieved a modest ROP improvement of 6% with a motorized RSS BHA, the second well achieved a more substantial 19% increase using a rotary steerable system. This progression highlights the adaptability and scalability of the AI solution across different BHA configurations and operational environments.

The AI system added value by providing real-time, data-driven recommendations that enabled the driller to optimize surface parameters such as Weight on Bit (WOB) and RPM. It also contributed to operational safety and tool longevity by detecting and mitigating dysfunctions such as stick-slip before they escalated.

Beyond pure ROP improvement, the AI system enhanced overall drilling efficiency, preserved downhole tool life, and reduced the likelihood of integrity-related non-productive time (NPT). These outcomes demonstrate that AI-driven optimization is not only a viable approach for performance enhancement but also a strategic enabler for cost-effective and reliable drilling operations.

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